

Use Measurement Confidently in Modelling and Design

Name: _____

Date: _____

Step 11

Upper Elementary



Today I am learning to use measurement confidently
to model, design and solve real-world problems.

1. Design a Playground

A playground area is:

- 12 m long
- 8 m wide



a) Draw a simple playground plan on the grid.



b) What is the total area?

Answer: _____ m²

c) A path 1 m wide surrounds the playground.
Explain how the area changes.

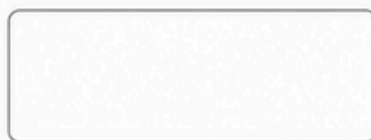
2. Build a Garden Box

A garden box measures:

- Length = 2.5 m
- Width = 1.5 m



a) Draw the garden box.



b) Which measurement tool would you use?

- ☐ Tape measure
☐ Ruler
☐ Clock



c) What units would be most appropriate?

- ☐ cm ☐ m ☐ km

3. Water Tank Design

A water tank holds:

- 2500 L

A second tank holds:

- 15 kL



a) Convert 15 kL into litres.

Answer: _____ L

b) Which tank holds more water?

c) How much more?

4. Science Investigation – Plant Growth Study



Week	Height
1	8 cm
2	11 cm
3	14 cm
4	17 cm

a) How much did the plant grow each week?

Answer: _____ cm

b) Predict the height at Week 6.

Answer: _____ cm

c) Explain your reasoning.

5. Measurement and Design Challenge

Design a bookshelf.

Requirements:

- Width = 1 m
- Height = 1.8 m
- Three shelves



a) Sketch your design.



b) Label measurements on your sketch.

c) Why did you choose those measurements?

6. Convert and Compare

a) 2.5 m = _____ cm

b) 350 cm = _____ m

c) 1.75 L = _____ mL

d) 4500 mL = _____ L

e) 3.2 kg = _____ g



Helpful Conversions

100 cm = 1 m

1000 mL = 1 L

1000 L = 1 kL

1000 g = 1 kg

7. Real-Life Modelling Problem

A class is creating a vegetable garden.

Available space:

- 8 m × 5 m

Garden beds (each):

- 2 m × 1 m



a) How many garden beds could fit?

Answer: _____

b) Show your thinking.

c) Draw a possible layout.



Scale: 1 square = 1 m

8. Think and Explain



a) Why is measurement important in design?



b) Why do designers use accurate measurements?



c) How does measurement help solve real-world problems?



I can...

- ☐ plan and design using measurement
☐ use the right units and tools
☐ convert and compare measurements
☐ solve real-world design problems

Measurement Words

length



width



height



area



volume



accuracy



design



Great job!

You are becoming
a confident and
creative problem solver!